MATH 142: Elementary Algebraic Topology

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11/9/26 Lecture 31: Simplicial Homology

Recall: If K is a simplicial complex,

$$C_q(K) = \left\{\sum n_i\sigma_i \,:\, n_i \in \mathbb{Z},\ \sigma_i \text{ an oriented } q\text{-simplex}\right\} / \{\tau + \sigma \,:\, \tau, \sigma \text{ same simplex with opposite orientation}\}$$

These are called the q-chains of K, and they form a free abelian group. Also recall

$$\partial : C_q(K) \to C_{q-1}(K)$$

by

$$\partial(v_0, \dots, v_q) = \sum (-1)^i (v_0, \dots, \widehat{v_i}, \dots, v_q) \in C_{q-1}(K)$$

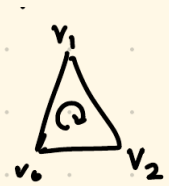
and

$$Z_q = \ker(\partial : C_q(K) \to C_{q-1}(K)) \quad \text{and} \quad B_q = \operatorname{Im}(\partial : C_{q+1}(K) \to C_q(K))$$

Question: What happens if you apply ∂ twice?

Example: A Triangle

Let σ be the oriented triangle

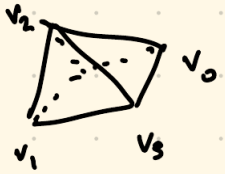


$$\partial(\sigma) = (v_0v_1) + (v_1v_2) + (v_2v_0) \in C_1(K)$$

It turns out, by computation, $\partial(\partial(\sigma)) = 0$.

Example: A Tetrahedron

Let σ be the tetrahedron



Take the orientation $(v_0v_1v_2v_3)$.

$$\partial\sigma = (v_1v_2v_3) - (v_0v_2v_3) + (v_0v_1v_3) - (v_0v_1v_2) \in C_2(K)$$

With the induced orientation, we can compute to see

$$\partial(\partial\sigma) = 0 \in C_1(K)$$

Lemma 31.1

$\partial^2 = 0$. That is,

$$C_{q+1}(K) \xrightarrow{\partial} C_q(K) \xrightarrow{\partial} C_{q-1}(K), \quad \partial \circ \partial : C_{q+1}(K) \to C_{q-1}(K) \text{ is } 0$$

Proof: It is enough to look at generators. Thus, consider $\partial(\partial\sigma)$ for σ a $(q+1)$-simplex. We see

$$\partial(v_0 v_1 \cdots v_{q+1}) = \sum (-1)^i (v_0, \ldots, \widehat{v_i}, \ldots, v_{q+1})$$

so

$$\begin{aligned}
\partial(\partial(v_0 v_1 \cdots v_{q+1})) &= \sum (-1)^i \partial(v_0, \ldots, \widehat{v_i}, \ldots, v_{q+1}) \\
&= \sum (-1)^i \sum_{j>i} (-1)^{j-1} (v_0, \ldots, \widehat{v_i}, \ldots, \widehat{v_j}, \ldots, v_{q+1}) \\
&\quad + \sum (-1)^i \sum_{j<i} (-1)^{j} (v_0, \ldots, \widehat{v_j}, \ldots, \widehat{v_i}, \ldots, v_{q+1}) \\
&= \sum_{k<\ell} (-1)^{k+\ell-1} (v_0, \ldots, \widehat{v_k}, \ldots, \widehat{v_\ell}, \ldots, v_{q+1}) \\
&\quad + \sum_{k<\ell} (-1)^{k+\ell} (v_0, \ldots, \widehat{v_k}, \ldots, \widehat{v_\ell}, \ldots, v_{q+1}) \\
&= 0
\end{aligned}$$

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Corollary 31.2

$B_q(K) \subseteq Z_q(K)$

Proof: $\operatorname{Im}(\partial : C_{q+1}(K) \to C_q(K)) \subseteq \ker(\partial : C_q(K) \to C_{q-1}(K))$ because $\partial^2 = 0$.

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Question: How different are $B_q(K)$ and $Z_q(K)$? This motivates the following definition.

Definition 31.3: Homology Groups

Let the qth *homology group* of K be

$$H_q(K) = Z_q(K)/B_q(K)$$

Definition 31.4: Homologous Cycles

If $z \in Z_q(K)$ is a cycle, write $[z]$ for its class in $H_q(K)$. If two cycles differ by a boundary, they are *homologous*.

Now, $H_q(K)$ is a finitely generated abelian group since K is finite (so $Z_q(K)$ and $B_q(K)$ are finitely generated).

Let

$$T_q(K) := \{\alpha \in H_q(K) \,:\, n\alpha = 0 \text{ for some } n \neq 0\}$$

a finite group, called the *q-torsion group* of K. Also, let $F_q(K) = H_q(K)/T_q(K)$, a free abelian group. This is isomorphic to some $\mathbb{Z}^{\beta_q}$ for $\beta_q \geq 0$. We call β_q the qth *Betti number* of K.

Note: $H_q(K) \cong T_q(K) \oplus \mathbb{Z}^{\beta_q(K)}$

Example: Boundary of a Triangle

Let K be the triangle with no interior, and let's compute its homology.



$$C_0(K) = \{n_0v_0 + n_1v_1 + n_2v_2\}$$
$$C_1(K) = \{n_2(v_0v_1) + n_0(v_1v_2) + n_1(v_2v_0)\}$$

Now, $\partial : C_0(K) \to 0$ since $C_{-1}(K) = 0$. Also,

$$\begin{aligned}\partial(n_2(v_0v_1) + n_0(v_1v_2) + n_1(v_2v_0)) &= n_2((v_1) - (v_0)) + n_0((v_2) - (v_1)) + n_1((v_0) - (v_2)) \\ &= (n_1 - n_2)(v_0) + (n_2 - n_0)(v_1) + (n_0 - n_1)(v_2) \\ &\in C_0(K)\end{aligned}$$

so

$$\begin{aligned}Z_0(K) &= C_0(K) \\ Z_1(K) &= \{z((v_0v_1) + (v_1v_2) + (v_2v_0)) \,:\, z \in \mathbb{Z}\} = \mathbb{Z}((v_0v_1) + (v_1v_2) + (v_2v_0)) \\ B_1(K) &= \operatorname{Im}(\partial : \underbrace{C_2(K)}_{=0} \to C_1(K)) = 0\end{aligned}$$

$$B_0(K) = \operatorname{Im}(\partial : C_1(K) \to C_0(K)) = \{m_0v_0 + m_1v_1 + m_2v_2 \,:\, m_0 + m_1 + m_2 = 0\}$$

so

$$\begin{aligned}H_1(K) &= Z_1(K)/B_1(K) \cong \mathbb{Z} \\ H_0(K) &= Z_0(K)/B_0(K) = C_0(K)/\{m_0v_0 + m_1v_1 + m_2v_2 \,:\, m_0 + m_1 + m_2 = 0\}\end{aligned}$$

To understand this, consider $\alpha : C_0(K) \to \mathbb{Z}$ by $\alpha(m_0v_0 + m_1v_1 + m_2v_2) = m_0 + m_1 + m_2$. Since $\ker(\alpha) = B_0(K)$ and α is surjective,

$$Z_0(K)/B_0(K) \cong \operatorname{Im}(\alpha) = \mathbb{Z}$$

Example: A Solid Triangle

Let K be the 2-dimensional simplex (the triangle with its interior).

We stated before $\alpha = (v_0v_1) + (v_1v_2) + (v_2v_0) \in Z_1(K)$ is still a 1-cycle, but this is the boundary of $(v_0v_1v_2)$. Thus, $[\alpha] = 0$ in $H_1(K)$.

By similar work, $H_0(K) = \mathbb{Z}$ and $H_i(K) = 0$ if $i > 0$.

Theorem 31.5

$H_0(K) = \mathbb{Z}^r$, where r is the number of components of K.

Proof: $Z_0(K) = \ker(\partial : C_0(K) \to 0) = C_0(K)$. Say that the components of $|K|$ are $C_1, \dots, C_r$.



Let $\phi : Z_0(K) \to \mathbb{Z}^r$ be

$$\phi\left(\sum n_i v_i\right) = (\text{sum of } n_i \text{ for } v_i \text{ in } C_1, \dots, \text{sum of } n_i \text{ for } v_i \text{ in } C_r)$$

We see ϕ is onto. Then, $z \in \ker(\phi)$ iff in every component C_j, the sum of the n_i's for vertices v_i in C_j is zero.

Take $z = \sum n_i v_i \in \ker(\phi)$. Choose a vertex v_j in C_j. Say P_i is an edge path from v_j to v_i in C_j. Let

$$\alpha = \sum_j \sum_{v_i \in C_j} n_i P_i \in C_1(K)$$

Then, as $\partial P_i = ((v_i) - (v_j))$ as seen before,

$$\begin{aligned}
\partial\alpha &= \sum_{j=1}^{r} \sum_{v_i \in C_j} n_i((v_i) - (v_j)) \\
&= \sum_{j=1}^{r} \sum_{v_i \in C_j} n_i v_i - \sum_{j=1}^{r} \underbrace{\left(\sum_{v_i \in C_j} n_i\right)}_{=0} v_j \\
&= \sum_{j=1}^{r} \sum_{v_i \in C_j} n_i v_i \\
&= z
\end{aligned}$$

for $z = \partial\alpha$. Thus, $z \in B_0$. It is also clear $B_0 \subseteq \ker(\phi)$, so

$$H_0(K) = Z_0(K)/B_0(K) = Z_0(K)/\ker\phi \cong \mathbb{Z}^r$$

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